12TH BIOLOGY · CHAPTER 17 · SOLVED EXERCISE

Human Reproductive Systems

Complete textbook exercise: MCQs, Short Questions, Long Questions and Inquisitive Questions

SECTION 1: MULTIPLE CHOICE QUESTIONS

1. Fertilization of the ovum occurs in:

(a) Oviduct (b) Uterus
(c) Cervix (d) Vagina

Answer: (a) Oviduct — The ovum is fertilized in the oviduct (fallopian tube) while it travels toward the uterus.

2. Which cells of the testis secrete testosterone hormone?

(a) Sertoli cells (b) Germline cells
(c) Spermatogonia (d) Leydig cells

Answer: (d) Leydig cells — LH stimulates the Leydig cells, located between the seminiferous tubules, to secrete testosterone.

3. The site of the production of sperm is:

(a) Seminiferous tubules (b) Epididymis
(c) Seminal vesicle (d) Vas deference

Answer: (a) Seminiferous tubules — Sperms are formed by the germline cells in the walls of the seminiferous tubules.

4. Which part produces fructose-rich fluid for the sperm?

(a) Prostate gland (b) Seminal vesicles
(c) Testis (d) Bulbourethral gland

Answer: (b) Seminal vesicles — The seminal vesicles release a fluid rich in fructose, which provides energy for the sperm.

5. In males, which hormone initiates spermatogenesis?

(a) Luteinizing hormone (b) Follicle stimulating hormone
(c) Gonadotropin-releasing hormone (d) Testosterone

Answer: (b) Follicle stimulating hormone — FSH stimulates the seminiferous tubules for spermatogenesis.

In the textbook, the label of the second option is printed as "b)" without the opening bracket.

6. During ovulation which cell is actually released from ovary?

(a) Oogonium (b) Primary oocyte
(c) Secondary oocyte (d) Mature ovum

Answer: (c) Secondary oocyte — The secondary oocyte, stopped in metaphase-II, is released. It becomes an ovum only after a sperm fuses with it.

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7. Which hormone stimulates the Graafian follicle to release ovum?

(a) LH (b) FSH
(c) Estrogen (d) Progesterone

Answer: (a) LH — The sudden release of LH on day 14 causes ovulation.

SECTION 2: SHORT QUESTIONS

1. Briefly describe the function of seminal vesicles, prostate gland, and bulbourethral glands.

1. **Seminal vesicles:** There are two seminal vesicles. Each releases a fluid rich in fructose (sugar) into the vas deferens through a duct. This fluid provides energy for the sperm.
2. **Prostate gland:** It lies just below the urinary bladder. It is soft and spongy and produces about 60% of the fluid of the semen.
3. **Bulbourethral glands:** There are two small bulbourethral glands. They secrete a clear fluid that lubricates the urethra for sperm passage and provides a protective neutral environment for sperms to survive.

2. When do the ova in a females begin meiosis? When do they complete the first meiotic division?

The primary oocytes begin the first meiotic division during the **embryonic development** of the female, but they stop in prophase-I and remain there all through childhood (for at least 10 to 14 years). They complete the first meiotic division at **puberty**, when one primary oocyte resumes meiosis-I and produces a secondary oocyte and the first polar body.

3. Write a brief note on the role of FSH, LH and testosterone in man.

- **FSH** stimulates the seminiferous tubules for spermatogenesis. It also causes the Sertoli cells to release inhibin, which inhibits further FSH secretion.
- **LH** stimulates the Leydig cells to secrete testosterone.
- **Testosterone** stimulates the development of the germinal epithelium and maintains the male secondary sexual characteristics. Its release inhibits the release of LH and GnRH.

4. List the structures of female reproductive system.

1. Gonads: two ovaries
2. Two oviducts (fallopian tubes or uterine tubes)
3. Uterus (fundus, body and cervix)
4. Vagina

The oviducts, uterus and vagina are called the accessory organs of the female reproductive system.

5. What changes occur in ovulation and menstruation during gestation period.

During the gestation period (pregnancy), both ovulation and menstruation stop. When the ovulated oocyte is fertilized, the tiny embryo secretes human chorionic gonadotropin (hCG). This hormone maintains the corpus luteum. So the levels of estrogen and progesterone remain high. The endometrium is not shed, so menstruation is prevented. Progesterone also inhibits FSH secretion and suppresses ovulation, so no new follicle ripens.

6. What is the role of the corpus luteum in a menstrual cycle?

The corpus luteum is the yellow glandular structure formed from the ruptured Graafian follicle after ovulation. It secretes estrogen and progesterone, which make the endometrium more vascular and glandular. If fertilization does not occur, it degenerates. The levels of estrogen and progesterone then fall abruptly, and the endometrium is shed (menstruation).

7. Differentiate between:

• Primary and secondary spermatocytes

Primary spermatocytes	Secondary spermatocytes
They are formed by the division of spermatogonia.	They are formed from primary spermatocytes by meiosis-I.
They are diploid (2n).	They are haploid (1n).
Each divides by meiosis-I to form two secondary spermatocytes.	Each divides by meiosis-II to form two spermatids.

• Spermatogenesis and oogenesis

Spermatogenesis	Oogenesis
It is the production of sperms.	It is the production of the ovum.
It occurs in the seminiferous tubules of the testes.	It occurs in the follicles of the ovaries (completed in the oviduct).
One spermatogonium produces four spermatids (sperms).	One primary oocyte produces one ovum and polar bodies.
Meiosis proceeds without stopping.	Meiosis stops in prophase-I and again in metaphase-II.

• Sertoli cell and Leydig cell

Sertoli cell	Leydig cell
It is present in the wall (germinal epithelium) of the seminiferous tubules.	It is located in the interstitial tissue between the seminiferous tubules.
It nourishes the developing sperms.	It secretes testosterone.
It is stimulated by FSH and releases inhibin.	It is stimulated by LH.

• Spermatids and spermatozoa

Spermatids	Spermatozoa
They are formed from secondary spermatocytes by meiosis-II.	They are formed by the maturation of spermatids.
They are immature haploid cells.	They are mature, elongated male gametes.
They do not have a head, midpiece and tail.	They have a head (with acrosome), a midpiece and a tail.

• Primary oocyte and secondary oocyte

Primary oocyte	Secondary oocyte
It is formed from an oogonium by mitosis.	It is formed from the primary oocyte by meiosis-I.
It is diploid (2n).	It is haploid (1n).
It stops division in prophase-I.	It stops division in metaphase-II.
It is formed during embryonic development.	It is formed at puberty and is released in ovulation.

SECTION 3: LONG QUESTIONS

1. Describe the structure of human male reproductive system. Identify the function of each part.

1. Gonads (testes)

Two testes are located outside the abdomen in a skin pouch called the **scrotum**. Each testis is oval-shaped, about 3–5 cm long and 2–3 cm wide. A testis is composed of highly convoluted **seminiferous tubules**. The wall of the seminiferous tubules, i.e., the germinal epithelium, contains two types of cells: **germline cells**, which make sperms (spermatozoa) by meiosis, and **Sertoli cells**, which nourish the developing sperms.

2. Accessory ducts

From the seminiferous tubules, the spermatozoa enter a network of tubules called the **rete testis**. From here they enter a long, coiled tube called the **epididymis**. Here the spermatozoa are still non-motile. They remain in the epididymis for at least 18 hours. Then they become motile and enter another long tube, the **vas deferens**, which passes into the abdominal cavity via the inguinal canal.

The vas deferens from each testis joins with a duct from a seminal vesicle. After joining, the tube is called the **ejaculatory duct**. The two ejaculatory ducts pass through the prostate gland. Here they open into the **urethra**, which comes from the urinary bladder. The urethra opens out of the body through the penis.

3. Accessory glands

These are the glands that add fluids to sperms to form semen.

1. **Seminal vesicles:** There are two seminal vesicles. Each releases a fluid rich in fructose (sugar) into the vas deferens through a duct. This fluid provides energy for the sperm.
2. **Prostate gland:** It lies just below the urinary bladder. It is soft and spongy and produces about 60% of the fluid of the semen.
3. **Bulbourethral glands:** There are two small bulbourethral glands. They secrete a clear fluid that lubricates the urethra for sperm passage and provides a protective neutral environment for sperms to survive.

4. Copulatory organ (penis)

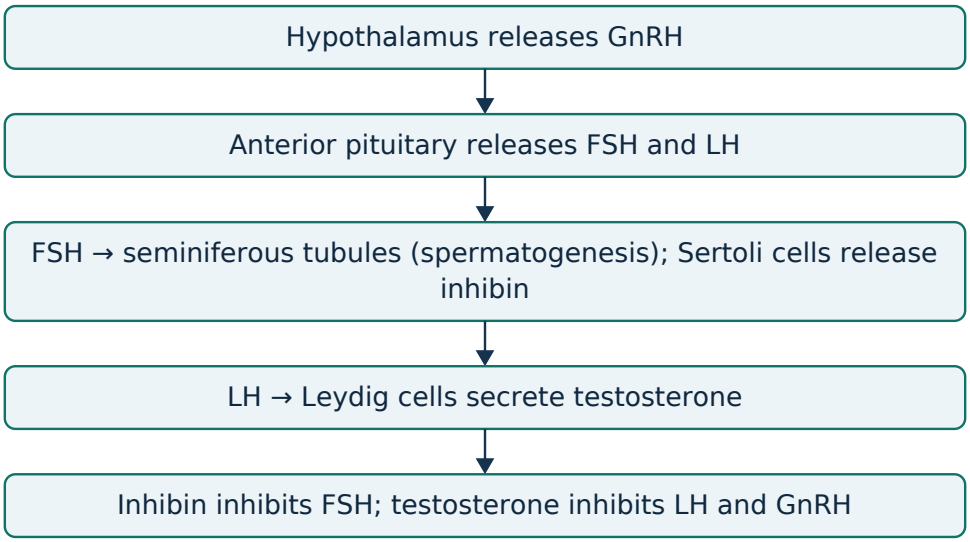
The male copulatory organ is called the penis. It contains the urethra, erectile tissue, an artery and two veins. It delivers the semen to the female reproductive system.

2. Explain the major reproductive hormones of human male and explain their role.

Spermatogenesis is controlled by the hormones of the hypothalamus and the pituitary gland.

- **GnRH:** The hypothalamus releases gonadotropin-releasing hormone (GnRH). It stimulates the anterior pituitary to release FSH and LH.
- **FSH:** Follicle-stimulating hormone stimulates the seminiferous tubules for spermatogenesis.
- **LH:** Luteinizing hormone stimulates the Leydig cells (located in the interstitial tissue between the seminiferous tubules) to secrete testosterone.
- **Testosterone:** It stimulates the development of the germinal epithelium and maintains the male secondary sexual characteristics.
- **Inhibin:** FSH causes the Sertoli cells to release a peptide hormone called inhibin, which inhibits further FSH secretion (negative feedback).
- Similarly, the release of testosterone inhibits the release of LH and GnRH (negative feedback).

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Hormonal control of the male reproductive system

3. Explain the structures of human female reproductive system. Identify the function of each part.

The female reproductive system consists of gonads, oviducts, uterus and vagina.

1. Gonads (ovaries)

Ovaries produce egg cells (ova or oocytes) and release hormones. There are two ovaries, each present on the side of the uterus. An oval-shaped ovary is 3–5 cm long and 2–3 cm wide. It contains many tiny sac-like structures called **ovarian follicles**, which are the functional units of the ovary. Follicles nurture immature eggs. In an adult female, one of the ripening follicles ejects its egg cell each month. This event is called **ovulation**. After ovulation, the ruptured follicle is transformed into a glandular structure called the **corpus luteum**.

2. Oviducts (fallopian tubes)

The oviducts (fallopian tubes or uterine tubes) are two tubes leading from the ovaries into the uterus. When an ovum matures, the ovarian follicle and ovary wall rupture to release it. The ovum enters an oviduct and travels toward the uterus. This movement is helped by the cilia on the inner lining of the oviduct. Fertilization of the ovum also occurs in the oviduct. The resulting zygote undergoes several mitotic divisions while still in the oviduct.

3. Uterus

The uterus is a hollow and muscular organ into which the two oviducts enter. It consists of three parts: the **fundus** (uppermost rounded portion above the openings of the oviducts), the **body** (major portion) and the **cervix** (lower narrow part). The opening of the cervix leads to the vagina.

The wall of the uterus has three layers. The **perimetrium** is the outermost thin layer. The **myometrium** is the middle thick layer of smooth muscles. The **endometrium** is the inner spongy lining. It has a basal layer and a functional layer. The basal layer has stem cells which regenerate the functional layer. The functional layer thickens and is shed during the menstrual cycle. During pregnancy, it thickens further to implant the

embryo and forms the **placenta**, which supplies oxygen and nutrition to the embryo and foetus.

4. Vagina

It is a thin-walled, 8–10 cm long tube leading from the outside to the cervix of the uterus. It provides a passageway for the delivery of the infant and for menstrual flow.

4. Describe the events of a menstrual cycle and explain its hormonal regulation.

The female reproductive system functions by going through cyclic events, collectively called the **menstrual cycle**. It lasts approximately one month (28 days on average). It has the following phases.

1. Menstrual phase (Days 1–5)

The functional layer of the endometrium is shed. The detached tissue and blood move out of the body. This is called **menstruation**. It continues for 3–5 days. It occurs due to low levels of estrogen and progesterone. FSH begins to rise in this phase.

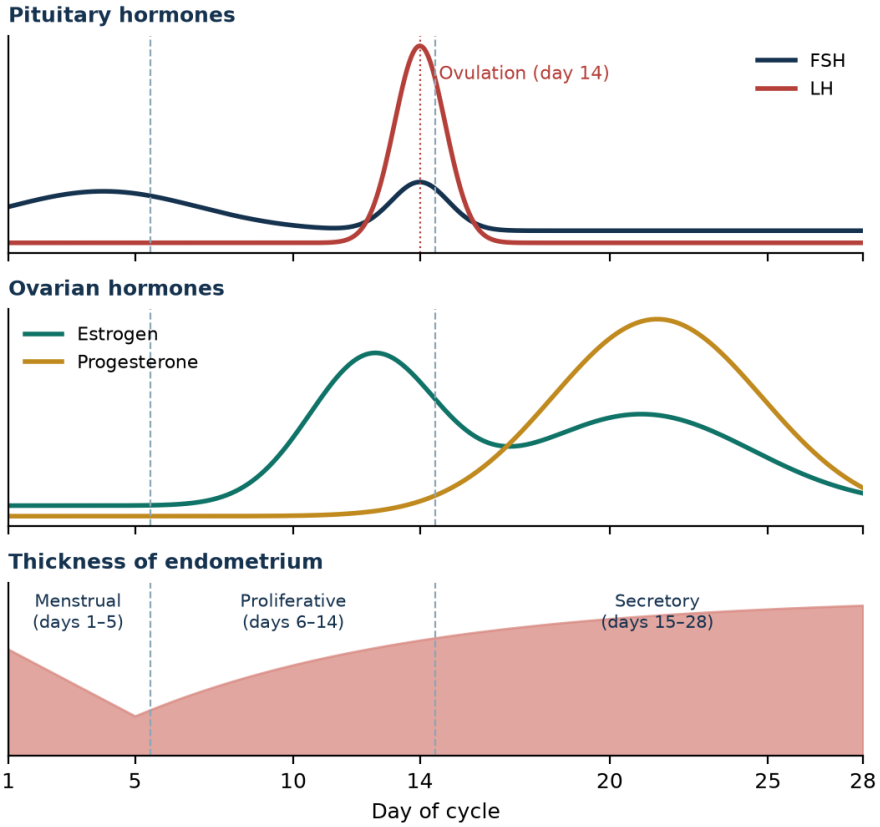
2. Proliferative phase and ovulation (Days 6–14)

Due to the rise in FSH, a few ovarian follicles begin growing, but only one grows fully and the others degenerate. This mature follicle is called the **Graafian follicle**. FSH stimulates it to secrete **estrogen**, which causes the initial vascularization of the endometrium and makes it velvety and thick again. As the concentration of estrogen rises, the level of FSH falls due to negative feedback, and the anterior pituitary releases **LH** at the end of this phase (day 14). The sudden release of LH stimulates the Graafian follicle to release the ovum into the oviduct. This is **ovulation**.

3. Secretory / post-ovulatory phase (Days 15–28)

After ovulation, LH stimulates the ruptured Graafian follicle to develop into a yellow structure called the **corpus luteum**. It secretes estrogen and **progesterone**, which make the endometrium more vascular and glandular.

If fertilization does not occur, the corpus luteum starts to degenerate and the anterior pituitary stops secreting LH. The destruction of the corpus luteum causes an abrupt fall in estrogen and progesterone. So the thickened endometrium is shed with bleeding, i.e., menstruation on day 28.



Hormone levels and endometrium during the menstrual cycle (schematic)

Summary of hormonal regulation

Hormone	Source	Role in the menstrual cycle
FSH	Anterior pituitary	Growth of ovarian follicles; stimulates the Graafian follicle to secrete estrogen
Estrogen	Graafian follicle (later corpus luteum)	Thickening and vascularization of the endometrium; lowers FSH by negative feedback
LH	Anterior pituitary	Causes ovulation (day 14); changes the ruptured follicle into the corpus luteum
Progesterone	Corpus luteum	Makes the endometrium more vascular and glandular; its fall causes menstruation

INQUISITIVE QUESTIONS

1. List the structures in order, through which a sperm passes on its way from the seminiferous tubules to the outside.

Seminiferous tubules → Rete testis → Epididymis → Vas deferens → Ejaculatory duct → Urethra (through the penis) → Outside.

2. Describe the effect of endocrine disrupting contaminants on the reproductive abilities.

Endocrine disrupting contaminants are chemicals in the environment (for example some pesticides and chemicals of plastics) which behave like the body's own hormones or block their action. Reproduction is controlled by hormones (GnRH, FSH, LH, testosterone, estrogen and progesterone) through feedback mechanisms. When these contaminants disturb this hormone balance, the following effects may occur:

- In males: low testosterone level, reduced production and quality of sperms, and infertility.
- In females: irregular menstrual cycle, failure of ovulation, and difficulty in becoming or remaining pregnant.
- In children: early or delayed puberty.

This topic is not discussed in the textbook; the answer is based on general knowledge.

3. Why are so many sperms produced in the male and so few ova produced in the female?

Many sperms: Sperms have to travel a long way through the female reproductive tract to reach the ovum in the oviduct, and most of them die on the way. So a very large number is needed to make sure that at least one sperm reaches and fertilizes the ovum. Also, one spermatogonium gives four sperms, and sperms are very small cells.

Few ova: In oogenesis, one primary oocyte gives only one large ovum; the other cells are small polar bodies. The ovum keeps most of the cytoplasm for the future embryo. Only one follicle ripens each month, because the female nourishes the developing baby inside her body and can normally support one baby at a time.

This is a reasoned answer; the textbook does not answer it directly.

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